

# GEETHANJALI V N

Chennai, India

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## Education

**Velammal Engineering College**

**September 2023 – May 2027**

*B.E. Computer Science and Engineering — CGPA: 7.87/10*

**Relevant Coursework:** Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks

## Technical Skills

**Languages:** Java, Python, SQL, JavaScript

**Web & Backend:** HTML, CSS, FastAPI, Flask, REST APIs

**Databases:** MySQL, MongoDB, DynamoDB, PostgreSQL, Redis

**Cloud & DevOps:** AWS (Lambda, API Gateway, EC2, S3, IAM), Docker, Git, GitHub, Agile, Scrum

**Core CS:** OOP, Operating Systems, Computer Networks, DBMS, System Design

**Competitive programming :** LeetCode | HackerRank | Skillrack

## Projects

**Research Paper Agent** | *LangGraph, Gemini, ChromaDB, FastAPI, Sentence Transformers*

**June 2026**

- Architected a multi-step LangGraph agentic pipeline that autonomously retrieves, ranks, and synthesizes 20+ academic papers per query from arXiv and Semantic Scholar, cutting manual literature review time by 70%.
- Implemented RAG pipeline with ChromaDB vector store and sentence-transformer embeddings; grounded all model responses in cited sources, eliminating hallucinated references.
- Automated literature review generation, methodology comparison, and research-gap identification — producing structured outputs (summaries, presentation outlines, insights) in under 60 seconds.
- Built REST API layer using FastAPI to serve agent queries with asynchronous execution, supporting concurrent research sessions.

**DocuMind AI** | *FastAPI, Gemini, OCR, PostgreSQL, AWS S3*

**April 2026**

- Developed an AI document intelligence platform processing 5+ document types (invoices, receipts, scanned PDFs) using Gemini multimodal LLM + Tesseract OCR, achieving 90%+ structured field extraction accuracy.
- Designed confidence-score-based human-review routing that reduced manual verification workload by 60% by auto-approving high-confidence extractions above a configurable threshold.
- Built REST APIs with FastAPI; persisted structured extraction results and raw documents to PostgreSQL and AWS S3, enabling full audit trail and re-processing capability.
- Implemented multimodal LLM pipeline converting unstructured PDFs and images to validated JSON records with schema enforcement and error recovery.

**URL Shortener** | *FastAPI, Redis, MySQL, Docker*

**December 2025**

- Engineered a high-throughput URL shortening service supporting custom aliases and handling 10k+ daily redirects, containerized via Docker for portable deployment.
- Achieved 40% reduction in p99 lookup latency by implementing a Redis TTL-based caching layer on top of MySQL, eliminating redundant database round-trips on hot URLs.
- Developed RESTful APIs for URL creation, retrieval, expiry management, and analytics tracking with per-URL click metrics and geographic data.

## Experience

**TAMIZHAN SKILLS**

**June 2025 – July 2025**

*IoT & Embedded Systems Intern*

*Remote*

- Built REST API endpoints to stream live sensor telemetry to a web dashboard, enabling real-time occupancy visualization across 20 parking slots.
- Diagnosed and resolved I2C/UART communication faults between microcontroller and sensor modules, improving data-packet success rate from 70% to 98%.

## Achievements & Certifications

- **Won First Prize – Blockathon'24, Velammal Engineering College** Aug 2024
- **Top 105 out of 10,000+ global submissions — Google On Campus Solution Challenge (Top 1%)** May 2025

**Portfolio:** <https://geethanjali-vn.vercel.app>

- **Artificial Intelligence Primer Certification – Infosys Springboard**

May 2026